e-inverted-pendulum rev A

Two-wheeled self-balancing robot. The PCB is the chassis: STM32WB55 (BLE), DRV8711 driving two MG310 gear motors through external half-bridges, MPU-6500 IMU, 3S LiPo. Firmware: github.com/embedded-pro/e-inverted-pendulum

1 Battery input and protection

XT30 3S LiPo input, 4 A fuse
AO4805 back-to-back P-FET switch:
reverse polarity, soft start, SW101
SMBJ15A clamp, VBAT sense divider

File: power_entry.kicad_sch

2 5V0 buck and 3V3 LDO

TPS62933F buck VBAT -> +5V, 1.2 MHz
EN UVLO: start 9.1 V, stop 8.5 V
TLV75533P LDO +5V -> +3V3, 500 mA

File: logic_power.kicad_sch

3 MCU, clocks, supplies, debug

STM32WB55RGV6 (BLE 5), pin map from
pin-data/pinmap.csv; core SMPS, HSE 32 MHz,
LSE 32.768 kHz, NRST / BOOT0 pads,
STDC14 SWD + virtual COM port

File: mcu.kicad_sch

4 2.4 GHz RF path and antenna

RF1 -> MLPF-WB55-01E3 filter / match
50 Ohm GCPW -> pi tuning (DNP shunts)
Printed IFA (TI SWRA117D)
DNP U.FL conducted test port

File: rf.kicad_sch

5 Motor drive (DRV8711 + external FETs)

DRV8711, PWMMODE 1 (indexer bypass)
Four CSD88537ND half-bridges, SS34 clamps
Rsense 50 mOhm + Kelvin taps, trip 2.0 A
VM bulk 220 uF, charge pump, pull-ups

File: drive.kicad_sch

6 Motor connectors and encoder inputs

Two JST PH 6-pin (MG310 motor + Hall A/B)
ESD -> 100 Ohm + 1 nF -> SN74LVC3G17
Encoder supply 5V0 (or 3V3) via 100 mA PTC
Third buffer input grounded (no index)

File: motors_encoders.kicad_sch

7 IMU and SPI flash

MPU-6500 6-axis IMU on SPI2, INT to PC6
W25Q16JV 16 Mbit flash on SPI1
(shared with the DRV8711)

File: imu_flash.kicad_sch

8 User interface, expansion, test points

Status LED (PB0) + 12 RGB LEDs (PA7)
Qwiic I2C3 + ESD, M24C64 EEPROM
ST25DV04KC NFC tag + printed coil
No buttons, no buzzer (D-035)

File: ui.kicad_sch

Power flow: 3S LiPo (9.0-12.6 V) -> XT30 J101 -> 4 A fuse F101 -> back-to-back P-FET switch Q101 (reverse polarity, soft start, SW101) -> VBAT -> SMBJ15A clamp -> DRV8711 + four CSD88537ND half-bridges -> MG310 motors
VBAT -> TPS62933F buck -> +5V (encoders, buzzer, LDO) -> TLV75533P LDO -> +3V3 (MCU, IMU, flash, I2C devices)
MCU core: +3V3 -> internal SMPS (L301 + L302) -> VFBSMPS; VDDA through ferrite FB301

Ground: one GND, L2 solid and never cut. The motor current returns through Rsense R509 / R512 at the bridges; the Kelvin taps NT501 / NT502 keep the sense lines off the power ground. Numbering: sheet n uses references n01-n99.
Sources for every value: docs/design-calcs.md, docs/decisions.md, simulation/REPORT.md.

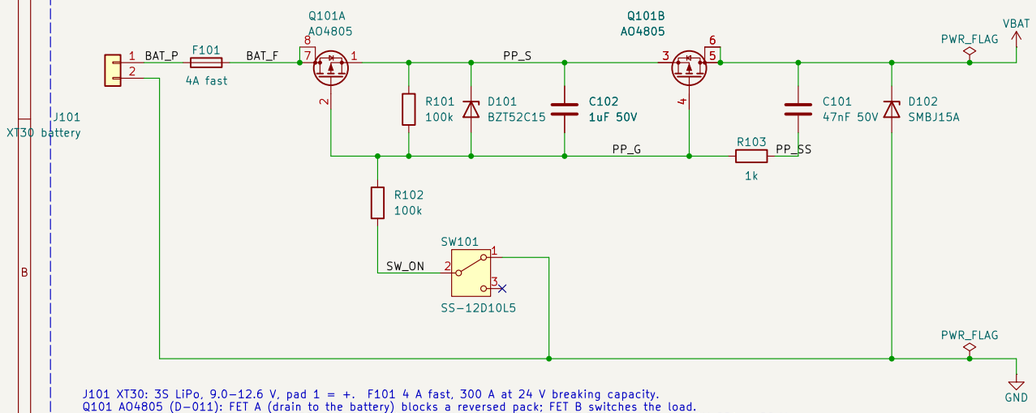
Generated by scripts/schgen/build.py (D-020): do not edit by hand
embedded-pro

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File: e-inverted-pendulum.kicad_sch

Title: e-inverted-pendulum: self-balancing robot chassis PCB

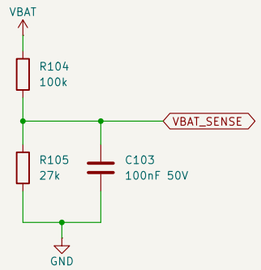
Size: A3	Date: 2026-11-13	**Rev: A**
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Input, fuse and power switch (D-011, D-021)



J101 XT30: 3S LiPo, 9.0–12.6 V, pad 1 = +. F101 4 A fast, 300 A at 24 V breaking capacity.
Q101 AO4805 (D-011): FET A (drain to the battery) blocks a reversed pack; FET B switches the load.
SW101 on: R102 pulls PP_G low, V_GS = −VBAT/2 (−4.5 V at 9.0 V, −6.8 V at 13.5 V): D101 keeps |V_GS| < 20 V.
SW101 off: R101 holds the gates off; C102 limits the hot-plug V_GS step to −0.57 V (< |V_th| 1.0 V) (S1).
Soft start (D-021): R103 + C101 Miller integrator, ~1.7 V/ms, inrush ~0.7 A for ~7 ms (S1); turn-off tau 105 ms.
D102 SMBJ15A after the FETs: V_RWM 15 V > 13.5 V, V_C 24.4 V < 30 V (AO4805) and 32 V (TPS62933).
It must never sit before the FETs: a reversed pack would forward-bias it through the fuse.

VBAT sense (ADC PC1)



VBAT_SENSE = VBAT x 27k / 127k = 0.2126 VBAT
(12.6 V -> 2.68 V, 13.5 V -> 2.87 V < VDDA).
Source 21.3 kOhm < 50 kOhm R_AIN max; C103 100 nF
>> 20 x 5 pF C_ADC, tau 2.1 ms (DS11929 Table 73).
+/-1.57 % ratio error with 1 % parts:
calibrate per board against VREFINT.

Test points



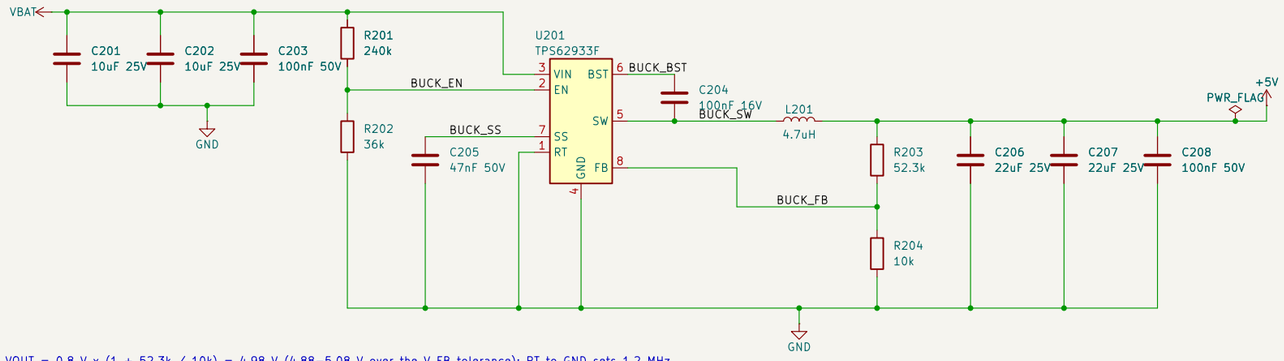
Generated by scripts/schgen/build.py (D-020): do not edit by hand
e-inverted-pendulum: self-balancing robot chassis PCB
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Sheet: /1 Battery input and protection/
File: power_entry.kicad_sch

Title: 1 Battery input and protection

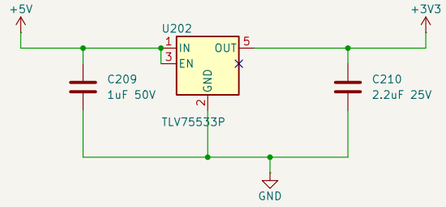
Size: A3	Date: 2026-11-13	**Rev: A**
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5V0 buck (TPS62933F, 1.2 MHz, forced PWM)



VOUT = 0.8 V x (1 + 52.3k / 10k) = 4.98 V (4.88–5.08 V over the V_FB tolerance); RT to GND sets 1.2 MHz.
L201 4.7 uH (I_sat 3.2 A): ripple 0.39 A at 9 V, 0.53 A at 12.6 V. C_OUT 2 x 22 uF + 100 nF (~22 uF at 5 V).
EN UVLO R201 / R202: start 9.11 V, stop 8.47 V (~2.8 V per cell protects the pack). C205 47 nF: t_SS 6.8 ms.
Layout: C203 at the VIN pin; minimal BUCK_SW copper, nothing under it on L3/L4; FB divider at the IC,
sensing at the output caps (docs/power-map.md).

3V3 LDO (TLV75533P)



TLV75533P: 3.3 V, 500 mA, EN tied to IN.
C_IN 1 uF, C_OUT 2.2 uF (>= 0.47 uF effective,
SBVS320A s6.3, s8.1.1).
0.20 W at 115 mA -> +45 C (R_thJA 231 C/W).

Test points



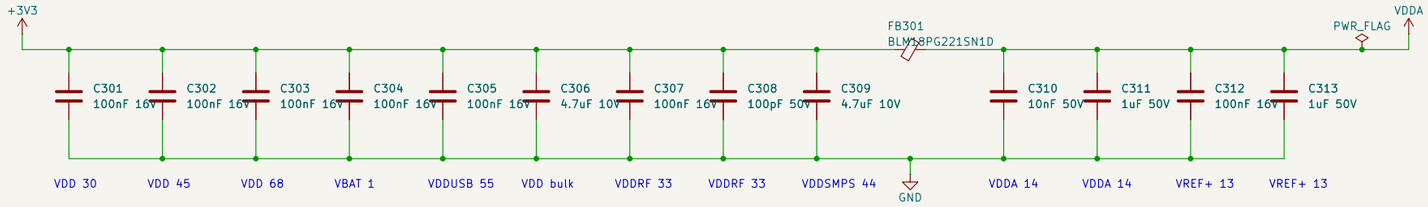
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e-inverted-pendulum: self-balancing robot chassis PCB
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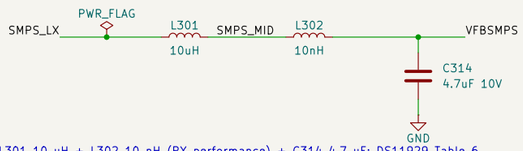
Size: A3	Date: 2026-11-13	**Rev: A**
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Supplies and decoupling (DS11929 Fig. 16)



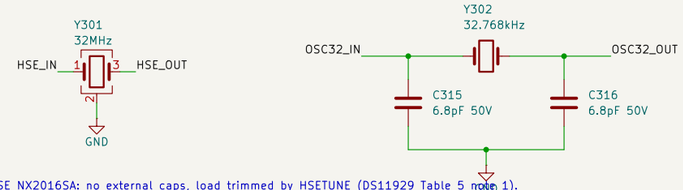
One 100 nF per supply pin, at the pin, each with its own via; 4.7 uF bulk on VDD. VDDRF 100 nF + 100 pF at pin 33.
VDDA and VREF+ through FB301: 10 nF + 1 uF on VDDA, 100 nF + 1 uF on VREF+ (mcu-circuit-design s2).
VDDUSB to VDD: USB unused (DS11929 s3.7). VBAT to VDD: no backup domain battery.

Core SMPS (DS11929 Table 6)



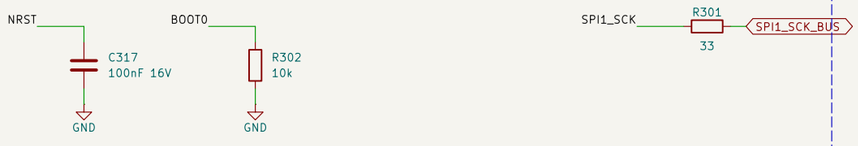
L301 10 uH + L302 10 nH (RX performance) + C314 4.7 uF: DS11929 Table 6.
L301 / L302 at pins 41-44, small loop, orthogonal to the RF line (docs/rf.md).

Clocks: HSE 32 MHz, LSE 32.768 kHz



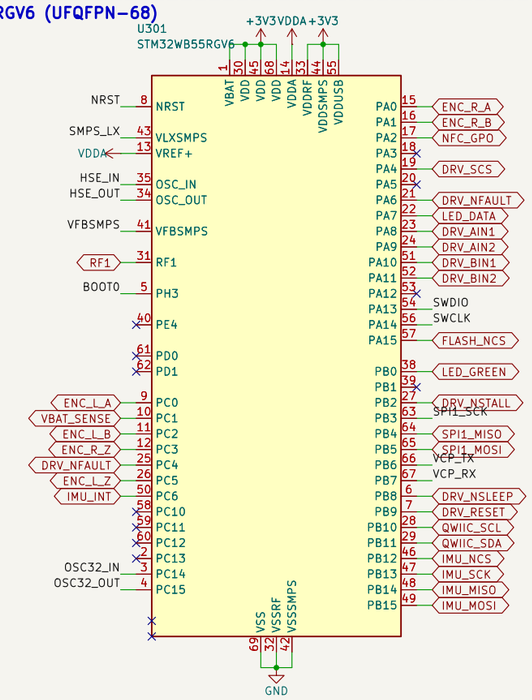
HSE NX2016SA: no external caps, load trimmed by HSETUNE (DS11929 Table 5 note 1).
LSE NX3215SA, C_L 6 pF: 2 x 6.8 pF C0G (AN2867): LSEDRV = 10, margin ~2.5 (Table 54).

Reset, boot and SPI1 clock termination



NRST: 100 nF at the pin (DS11929 Fig. 24); the STDC14 probe drives it too.
BOOT0 (PH3): 10k pull-down, no button (D-035): TP302 to +3V3 at reset, or the boot option bytes over SWD.
R301 33 Ohm at PB3: source termination of the SCK shared by the DRV8711 and the flash (s9).

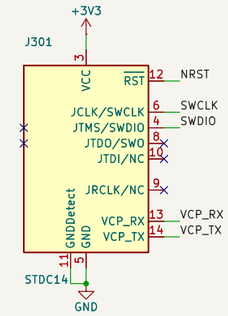
STM32WB55RGV6 (UFQFPN-68)



Test points



Debug and virtual COM port: STDC14 (D-009)



STLINK-V3 STDC14 (1.27 mm): SWD + virtual COM port on USART1 (PB6 TX, PB7 RX).
SWO (pin 8) unused: PB3 is SPI1_SCK. Pin 3 is the target VCC reference for the probe.

Pin plan (pin-data/pinmap.csv, checked by scripts/check_pinmap.py)

PA8	TIM1_CH1	DRV_AIN1	PC2	LPTIM1_IN2	ENC_L_B
PA9	TIM1_CH2	DRV_AIN2	PC5	EXTI5	ENC_L_Z
PA10	TIM1_CH3	DRV_BIN1	PA0	TIM2_CH1	ENC_R_A
PA11	TIM1_CH4	DRV_BIN2	PA1	TIM2_CH2	ENC_R_B
PA6	TIM1_BKIN	DRV_NFAULT	PC3	EXTI3	ENC_R_Z
PC4	EXTI4	DRV_NFAULT	PB6	USART1_TX	VCP_TX
PB3	SPI1_SCK	SPI1_SCK	PB7	USART1_RX	VCP_RX
PB4	SPI1_MISO	SPI1_MISO	PB0	GPIO	LED_GREEN
PB5	SPI1_MOSI	SPI1_MOSI	PA13	SYS_JTMS-SWDIO	SWDIO
PA4	GPIO	DRV_SCS	PA14	SYS_JTCK-SWCLK	SWCLK
PB8	GPIO	DRV_NSLEEP	PA15	GPIO	FLASH_NCS
PB9	GPIO	DRV_RESET	PC1	ADC1_IN2	VBAT_SENSE
PB13	SPI2_SCK	IMU_SCK	PA7	TIM17_CH1	LED_DATA
PB14	SPI2_MISO	IMU_MISO	PB10	I2C3_SCL	QWIIC_SCL
PB15	SPI2_MOSI	IMU_MOSI	PB11	I2C3_SDA	QWIIC_SDA
PB12	GPIO	IMU_NCS	PB2	GPIO	DRV_NSTALL
PC6	EXTI6	IMU_INT	PH3	BOOT0	BOOT0
PC0	LPTIM1_IN1	ENC_L_A	PA2	EXTI2	NFC_GPO

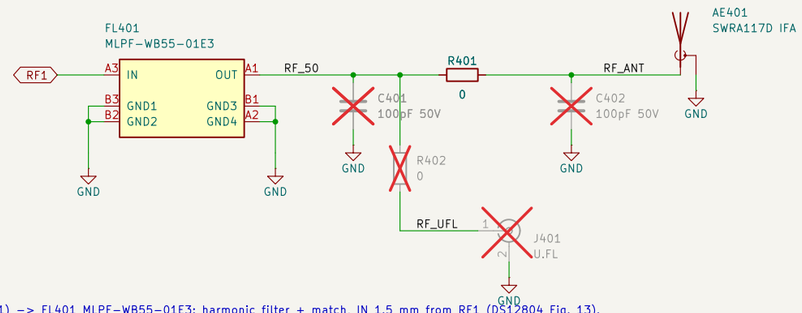
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e-inverted-pendulum: self-balancing robot chassis PCB
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File: mcu.kicad_sch

Title: 3 MCU, clocks, supplies, debug

Size: A3	Date: 2026-11-13	**Rev: A**
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2.4 GHz RF path (DS12804, D-007, docs/rf.md)



RF1 (pin 31) -> FL401 MLPF-WB55-01E3: harmonic filter + match, IN 1.5 mm from RF1 (DS12804 Fig. 13).
RF_50: 50 Ohm GCPW 0.27 / 0.15 mm on L1 over the solid L2 GND (scripts/gcpw.py, docs/rf.md).
Pi tuning: C401 / C402 DNP, R401 0R; tune with a VNA at the antenna feed (rf-layout s3).
Conducted test: fit R402, remove R401 -> J401 U.FL (DNP in production).
Layout: L1 only, no vias on the line, via fence 1.0-1.3 mm; antenna keep-out on all layers;
keep the SMPS (L301 / L302) away and orthogonal to the line.

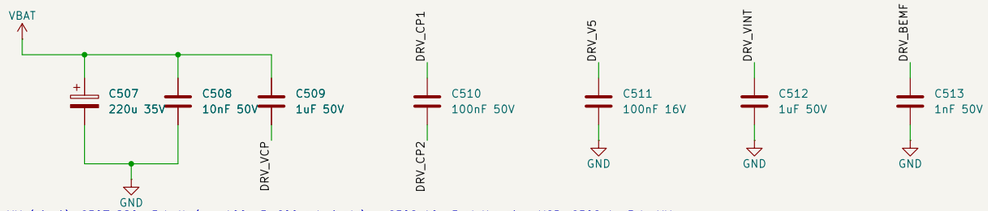
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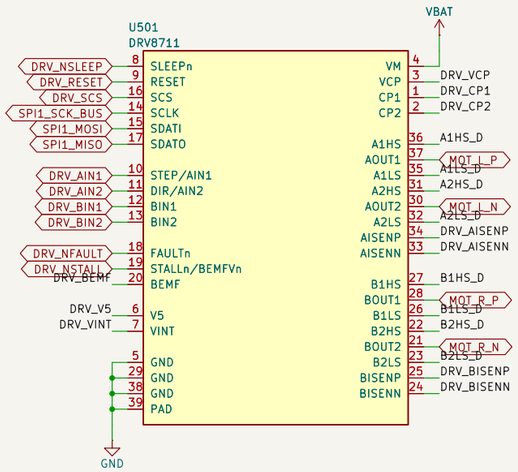
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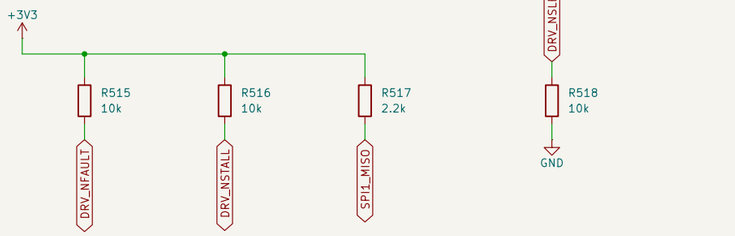
Size: A3	Date: 2026-11-13	**Rev: A**
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DRV8711 gate driver (PWMMODE 1, D-002)



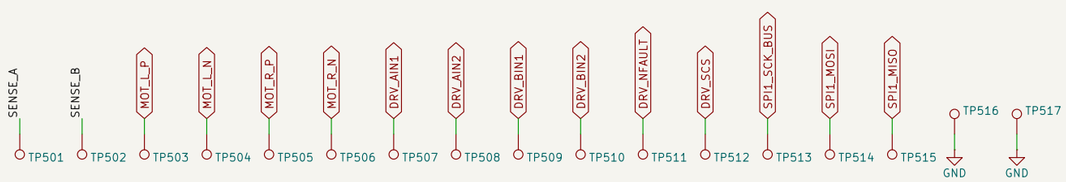
VM (pin 4): C507 220 uF bulk (>= 100 uF, 600 mA ripple) + C508 10 nF at the pin. VCP: C509 1 uF to VM.
CP1-CP2: C510 100 nF, >= VM + 12 V. V5 100 nF, VINT 1 uF, BEMF 1 nF COG (SLVSC40 p.3-4).



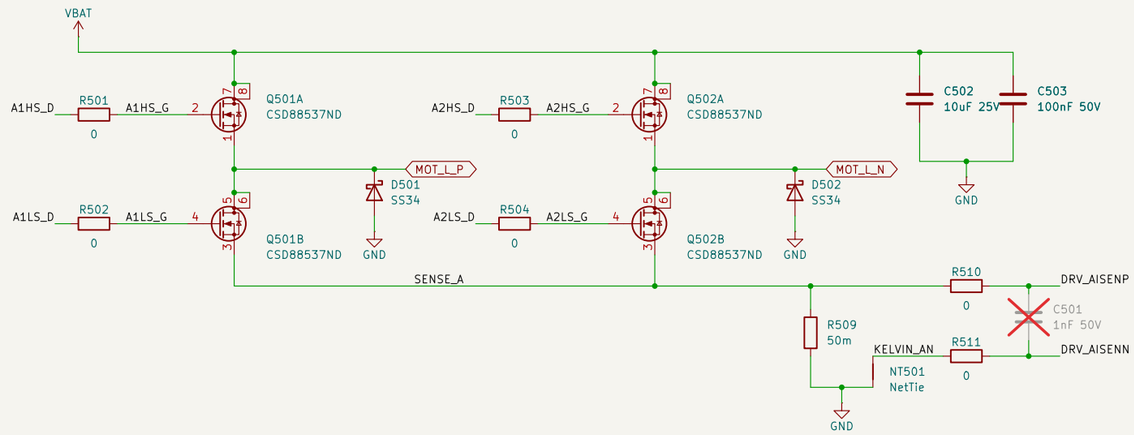


nFAULT, nSTALL: open drain, 10k to +3V3. SDATO is open drain on the shared MISO: 2.2k (tau ~66 ns, SPI <= 4 MHz).
nSLEEP: 10k pull-down, no internal pull, so the driver sleeps while the MCU is in reset (SLVSC40 p.3).
PWMMODE 1: TIM1 drives AIN1/AIN2/BIN1/BIN2 directly (indexer bypassed). Firmware: IDRIVE 00,
TBLANK >= 1.2 us, SCLK <= 4 MHz, SCS active high. Trip 2.0 A: ISGAIN 20, TORQUE 186 (D-027, S0).

Test points: motor current, phases, PWM, SPI

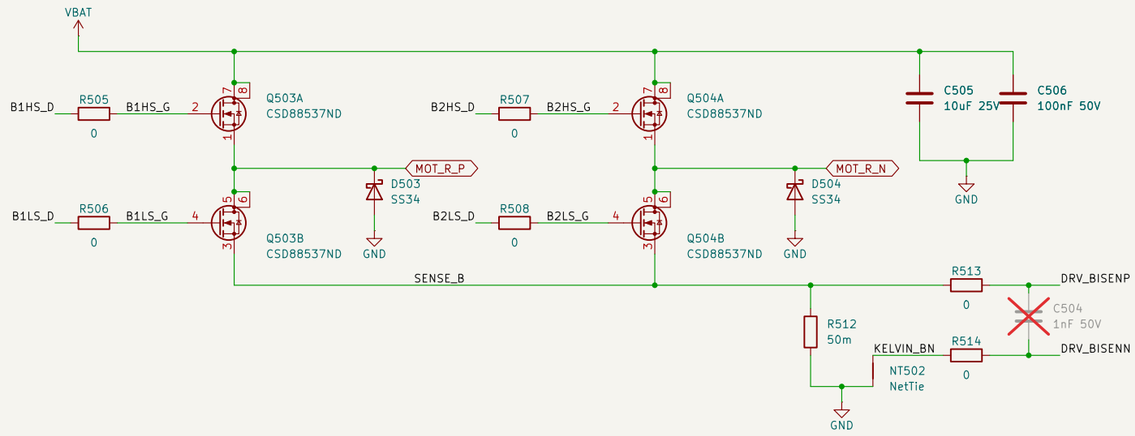


Bridge A: left MG310 (MOT_L_P / MOT_L_N)



A1 / A2: CSD88537ND, FET1 high side, FET2 low side; 0 Ohm gate R at each FET (edge-rate trim).
SS34 from GND to each phase node: freewheel at -0.36 V, not the -0.73 V body diode (phase pins abs max -0.6 V, S2).
R509 50 mOhm: the motor current flows straight through it; Kelvin taps from its inner pad edges (NT501 at the GND pad).
0 Ohm filter legs, C501 DNP: fit 27-33 Ohm + 1 nF if noisy (SLVSC40 Fig. 6). C502 + C503 at the HS drains: hot loop over L2 GND.

Bridge B: right MG310 (MOT_R_P / MOT_R_N)



B1 / B2: CSD88537ND, FET1 high side, FET2 low side; 0 Ohm gate R at each FET (edge-rate trim).
SS34 from GND to each phase node: freewheel at -0.36 V, not the -0.73 V body diode (phase pins abs max -0.6 V, S2).
R512 50 mOhm: the motor current flows straight through it; Kelvin taps from its inner pad edges (NT502 at the GND pad).
0 Ohm filter legs, C504 DNP: fit 27-33 Ohm + 1 nF if noisy (SLVSC40 Fig. 6). C505 + C506 at the HS drains: hot loop over L2 GND.

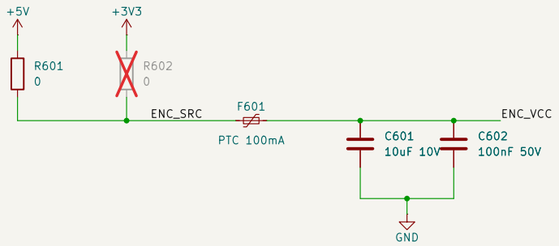
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e-inverted-pendulum: self-balancing robot chassis PCB
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Sheet: /5 Motor drive (DRV8711 + external FETs)/
File: drive.kicad_sch

Title: 5 Motor drive (DRV8711 + external FETs)

Size: A3	Date: 2026-11-13	**Rev: A**
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Encoder supply (D-016)

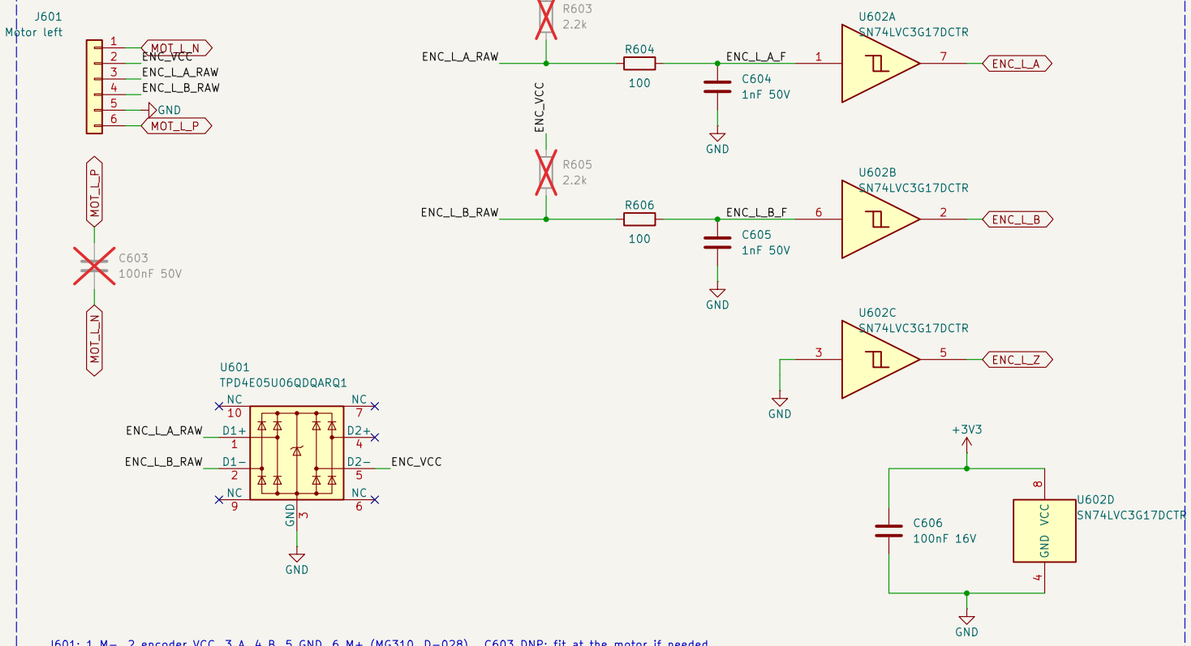


ENC_VCC: 5V0 through R601 (default, D-016)
or 3V3 through R602 (DNP).
F601 100 mA PTC: a shorted encoder cable
can't pull the rail down.

Encoder input chain (D-012, docs/design-calcs.md)

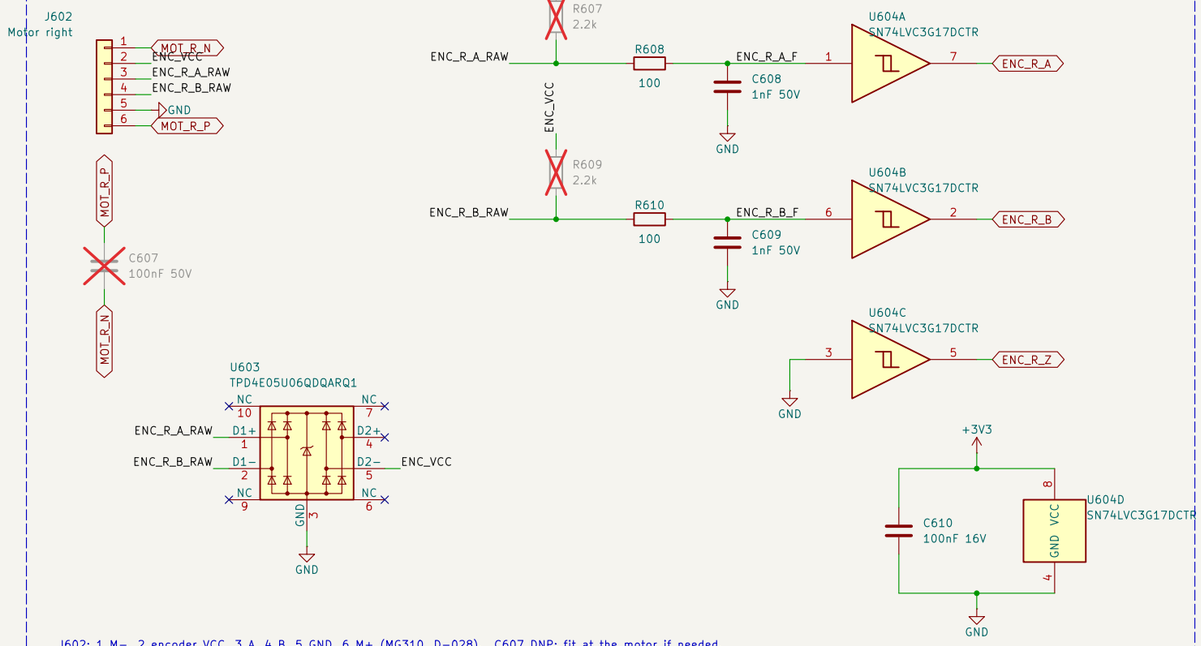
Per channel: ESD at the connector -> 2.2k open-collector pull-up (DNP, for open-collector encoders)
-> 100 Ohm + 1 nF: f_c 1.59 MHz > 2 x 105 kHz max channel rate, rise 220 ns
-> SN74LVC3G17 Schmitt buffer at 3V3: V_T+ <= 1.87 V, 5.5 V tolerant inputs with I_off (SCES470D)
-> LPTIM1 (left) / TIM2 (right) encoder inputs.
The MG310 has no index: the third input is grounded, PC5 / PC3 read low (D-028).
Layout: MOT_x away from every ENC_* trace on all layers (docs/power-map.md).

Left MG310: motor + Hall encoder (J601)



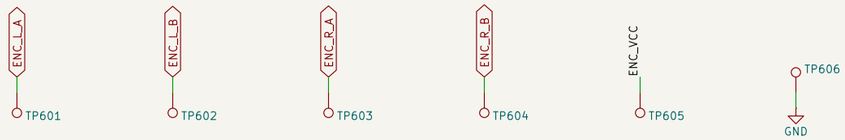
J601: 1 M-, 2 encoder VCC, 3 A, 4 B, 5 GND, 6 M+ (MG310, D-028). C603 DNP: fit at the motor if needed.
U601 at the connector, pass-through routing. The third buffer input is grounded: ENC_L_Z reads low.

Right MG310: motor + Hall encoder (J602)



J602: 1 M-, 2 encoder VCC, 3 A, 4 B, 5 GND, 6 M+ (MG310, D-028). C607 DNP: fit at the motor if needed.
U603 at the connector, pass-through routing. The third buffer input is grounded: ENC_R_Z reads low.

Test points: buffered encoder outputs



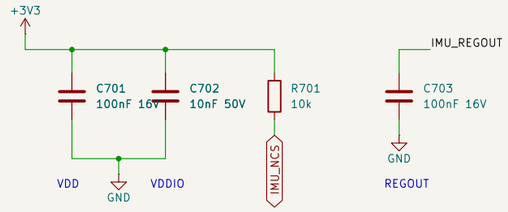
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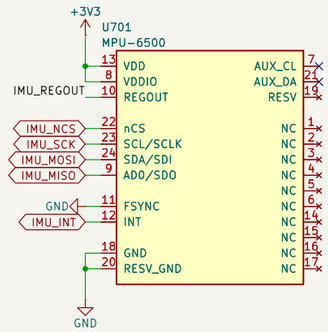
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Title: 6 Motor connectors and encoder inputs

Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/9

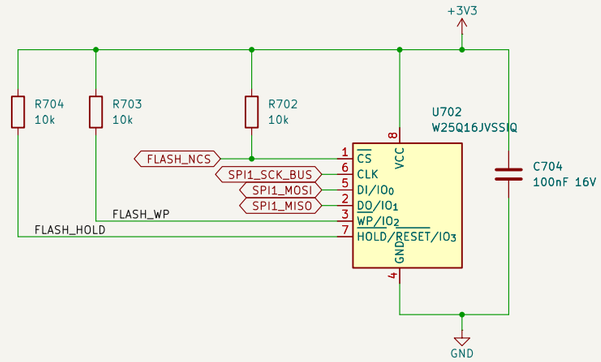
IMU: MPU-6500 on SPI2 (D-004)





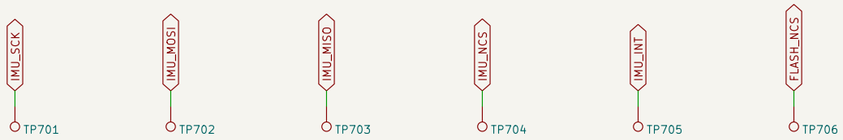
SPI2: PB13 SCK, PB15 MOSI, PB14 MISO, PB12 nCS (GPIO); INT to PC6 (EXTI6, data ready).
Decoupling per PS-MPU-6500A-01 Table 11: C1 100 nF REGOUT, C2 100 nF VDD, C3 10 nF VDDIO.
R701 holds nCS high until the MCU drives it; SDO stays high-Z and the part stays in SPI mode.
FSYNC unused, tied to GND (Table 10); AUX_DA / AUX_CL unconnected.
Mount rigidly beside the MCU, ~90 mm above the axle, axes x forward / y left / z up (HW-SNS-002, D-029).

SPI flash: W25Q16JV on SPI1 (D-005)



W25Q16JVSSIQ, 16 Mbit: SPI1 shared with the DRV8711 (SCK through R301 on the MCU sheet). /CS on PA15.
R702 keeps /CS at VCC through power-up (s10.3). /WP and /HOLD pulled up so IO2 / IO3 are defined;
the IQ variant ships with QE = 1.

Test points: IMU SPI2 and flash chip select



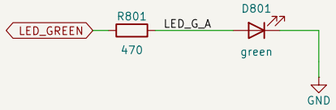
Generated by scripts/schgen/build.py (D-020): do not edit by hand
e-inverted-pendulum: self-balancing robot chassis PCB
embedded-pro

Sheet: /7 IMU and SPI flash/
File: imu_flash.kicad_sch

Title: 7 IMU and SPI flash

Size: A3	Date: 2026-11-13	**Rev: A**
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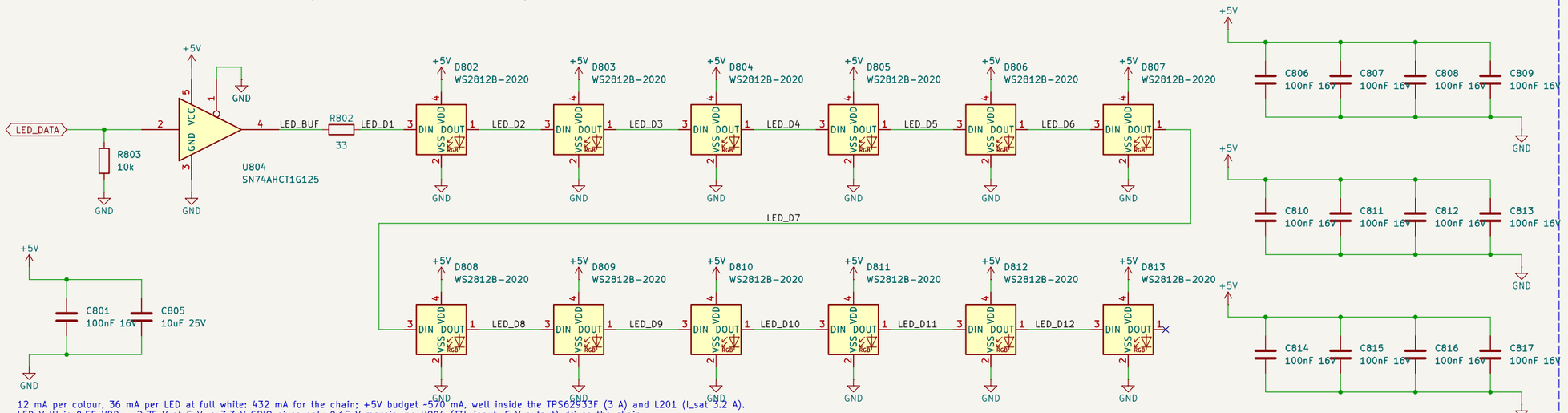
Status LED (PB0)



The firmware's statusLed (PlatformImpl, PB0) drives it: it stays 1:1 with the NUCLEO pin map.

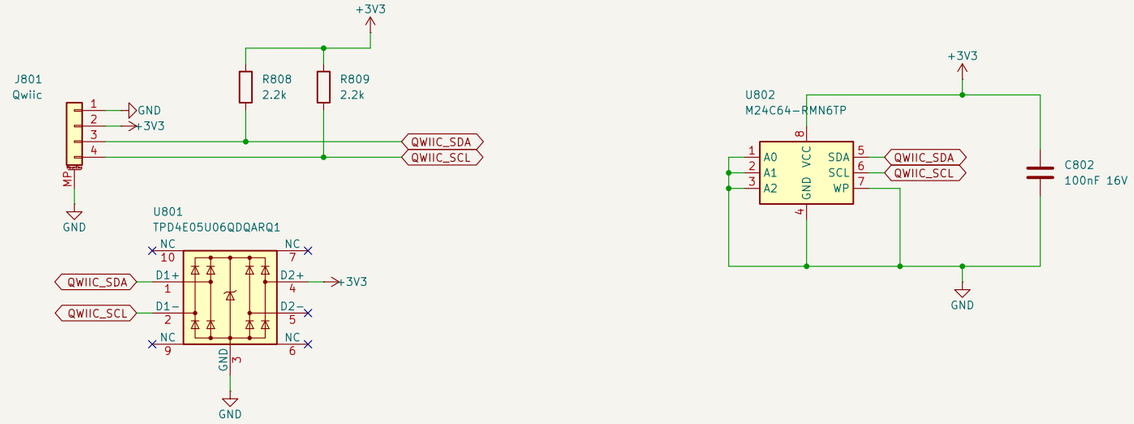
Green 470 Ohm: ~1.8 mA (V_F 2.45 V at 2 mA, KT-0603G p.5).

RGB LEDs: 12 x WS2812B-2020-V6 on +5V (PA7 TIM17_CH1 + DMA, D-035)



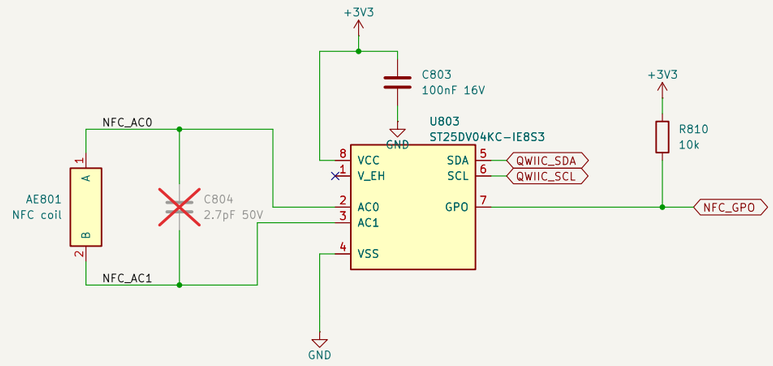
12 mA per colour, 36 mA per LED at full white: 432 mA for the chain; +5V budget ~570 mA, well inside the TPS62933F (3 A) and L201 (I_sat 3.2 A).
LED V_IH is 0.55 VDD = 2.75 V at 5 V; a 3.3 V GPIO gives only 0.15 V margin, so U804 (TTL input, 5 V output) drives the chain.
WS2812 protocol, GRB order, 800 kHz from TIM17 + DMA; reset >= 280 us. One 100 nF per LED at its VDD pin (C806-C817). Brightness is a firmware choice.

Qwiic I2C3 expansion and EEPROM (PB10 SCL, PB11 SDA)



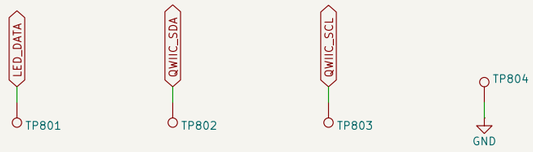
Qwiic pinout: 1 GND, 2 3V3, 3 SDA, 4 SCL. 2.2k pull-ups: window 967-3541 Ohm, Fast mode, C_b 100 pF (SLVA689A).
U801 ESD at the connector. U802 M24C64-R at 0x50: E2..E0 low, WC low = writes enabled (D-031).

NFC tag: ST25DV04KC + printed 13.56 MHz coil (D-030)



ST25DV04KC: ISO 15693 / NFC Type 5 tag on I2C3 (0x53 user, 0x57 system); V_EH unused; GPO open drain -> PA2 (EXTI2).
AE801: 10 turns, 25 x 21 mm, 4.25 uH; f0 14.2 MHz with C_TUN 28.5 pF + 1 pF (scripts/nfc_coil.py, docs/nfc.md).
C804 2.7 pF DNP across AC0-AC1 trims f0 to 13.56 MHz: fit after a phone / VNA check. Keep metal off the coil.

Test points



Generated by scripts/schgen/build.py (D-020): do not edit by hand
e-inverted-pendulum: self-balancing robot chassis PCB
embedded-pro

Sheet: /8 User interface, expansion, test points/		
File: ui.kicad_sch		
Title: 8 User interface, expansion, test points		
Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 9/9